

learn about the 555

by **Jimmy Proton** on December 23, 2010

Table of Contents

learn about the 555	1
Intro: Learn about the 555	2
Step 1: An overview of the 555 timer	2
Step 2: Pin configuration of the 555 timer	3
Step 3: Inside the 555 timer	3
Step 4: 555 timer operating modes	4
Step 5: Using the output of a 555 timer	5
Step 6: Common mistakes when using a 555 timer	6
Step 7: Thats it!	8
Related Instructables	8
Comments	8



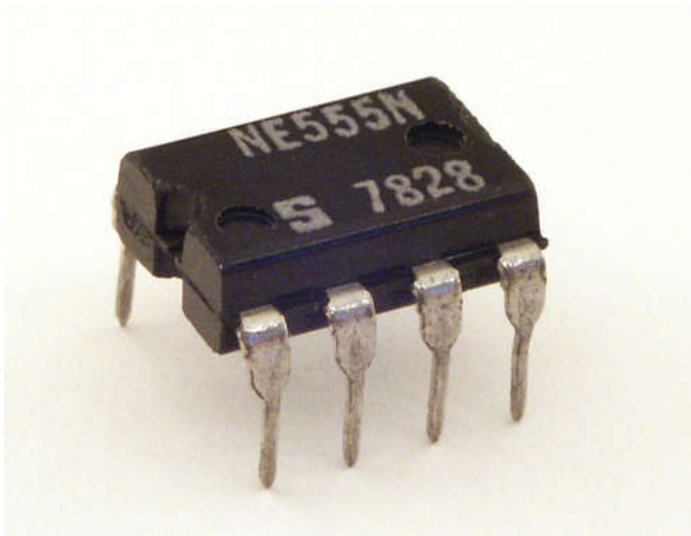
Author: Jimmy Proton

I am a Baptist Christian, I live in the USA, my dad was an electrical engineer in the navy, my uncle and grandpa are both engineers, I am interested in building technology.....
even when i clean it.....
parts.....
OOo! I like skinny girls! alot!

Intro: Learn about the 555

In this instructable i will teach you every thing you would ever need to know about the 555 timer IC. If you already know about the chip you could check out my slide show titled "47 projects to do with a 555" it will teach you every basic project to use a 555 with, its great for beginners!

I do not take 100% credit for this. source link on each step.



Step 1: An overview of the 555 timer

The 555 Integrated Circuit (IC) is an easy to use timer that has many applications. It is widely used in electronic circuits and this popularity means it is also very cheap to purchase, typically costing around 30p. A 'dual' version called the 556 is also available which includes two independent 555 ICs in one package.

For the 555 to function it relies on both analogue and digital electronic techniques, but if we consider its output only, it can be thought of as a digital device. The output can be in one of two states at any time, the first state is the 'low' state, which is 0v. The second state is the 'high' state, which is the voltage V_s (The voltage of your power supply which can be anything from 4.5 to 15v. 18v absolute maximum). The most common types of outputs can be categorized by the following (their names give you a clue as to their functions):

Monostable mode: in this mode, the 555 functions as a "one-shot". Applications include timers, missing pulse detection, bouncefree switches, touch switches, frequency divider, capacitance measurement, pulse-width modulation (PWM) etc

Astable - free running mode: the 555 can operate as an oscillator. Uses include LED and lamp flashers, pulse generation, logic clocks, tone generation, security alarms, pulse position modulation, etc.

Bistable mode or Schmitt trigger: the 555 can operate as a flip-flop, if the DIS pin is not connected and no capacitor is used. Uses include bouncefree latched switches, etc.

<http://www.555-timer-circuits.com/an-overview.html>

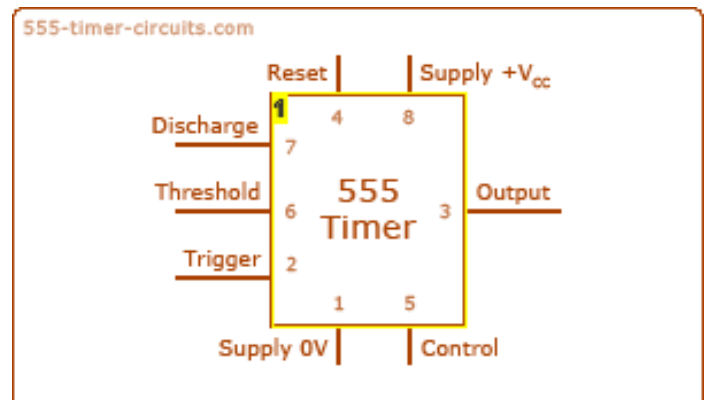
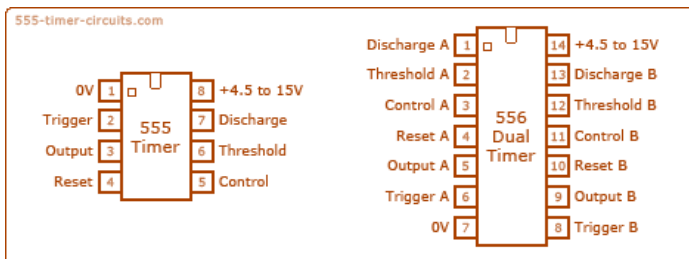


Image Notes

1. In a circuit diagram the 555 timer chip is often drawn like the illustration seen

<http://www.instructables.com/id/learn-about-the-555/>

here. Notice how the pins are not in the same order as the actual chip, this is because it is much easier to recognize the function of each pin, and makes drawing circuit diagrams much easier.

Step 2: Pin configuration of the 555 timer

This is the identification of each pin.

When drawing a circuit diagram, always draw the 555 as a building block, as shown in picture two with the pins in the following locations. This will help you instantly recognise the function of each pin:

Pin 1 (Ground):

Connects to the 0v power supply

Pin 2 (Trigger):

Detects 1/3 of rail voltage to make output HIGH. Pin 2 has control over pin 6. If pin 2 is LOW, and pin 6 LOW, output goes and stays HIGH. If pin 6 HIGH, and pin 2 goes LOW, output goes LOW while pin 2 LOW. This pin has a very high impedance (about 10M) and will trigger with about 1uA.

Pin 3 (Output):

(Pins 3 and 7 are "in phase.") Goes HIGH (about 2v less than rail) and LOW (about 0.5v less than 0v) and will deliver up to 200mA.

Pin 4 (Reset):

Internally connected HIGH via 100k. Must be taken below 0.8v to reset the chip.

Pin 5 (Control):

A voltage applied to this pin will vary the timing of the RC network (quite considerably).

Pin 6 (Threshold):

Detects 2/3 of rail voltage to make output LOW only if pin 2 is HIGH. This pin has a very high impedance (about 10M) and will trigger with about 0.2uA.

Pin 7 (Discharge):

Goes LOW when pin 6 detects 2/3 rail voltage but pin 2 must be HIGH. If pin 2 is HIGH, pin 6 can be HIGH or LOW and pin 7 remains LOW. Goes OPEN (HIGH) and stays HIGH when pin 2 detects 1/3 rail voltage (even as a LOW pulse) when pin 6 is LOW. (Pins 7 and 3 are "in phase.") Pin 7 is equal to pin 3 but pin 7 does not go high - it goes OPEN. But it goes LOW and will sink about 200mA.

Pin 8 (Supply):

Connects to the positive power supply (Vs). This can be any voltage between 4.5V and 15V DC, but is commonly 5V DC when working with digital ICs.

<http://www.555-timer-circuits.com/pin-configuration.html>

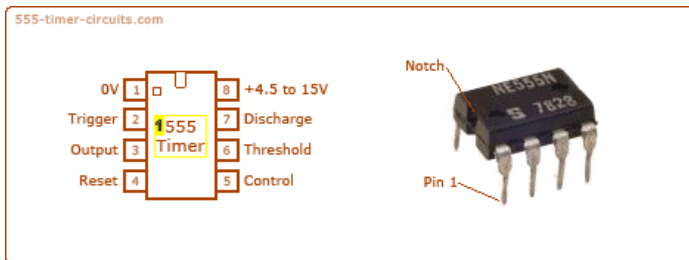


Image Notes

1. picture one

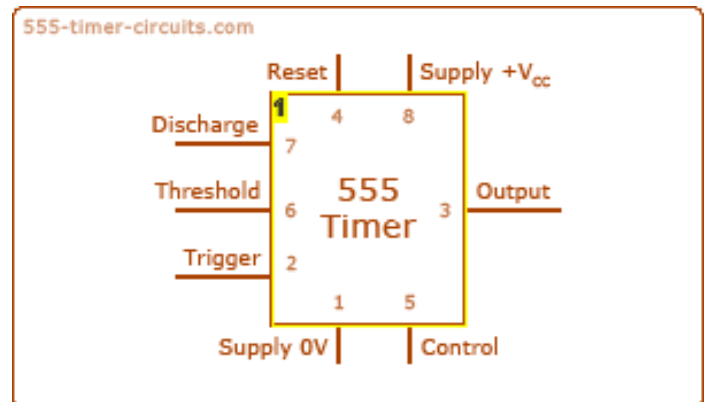


Image Notes

1. In a circuit diagram the 555 timer chip is often drawn like the illustration seen here. Notice how the pins are not in the same order as the actual chip, this is because it is much easier to recognize the function of each pin, and makes drawing circuit diagrams much easier.

Step 3: Inside the 555 timer

You may wonder what is inside the 555 timer chip or what makes it work. Well, the 555 timer chip is an Integrated Circuit (IC) and therefore it contains a miniaturized circuit surrounded by silicon. Each of the pins is connected to the circuit which consists of over 20 transistors, 2 diodes and 15 resistors.

Do you notice the three 5k resistors? This is how the chip got its name.

<http://www.555-timer-circuits.com/pin-configuration.html>

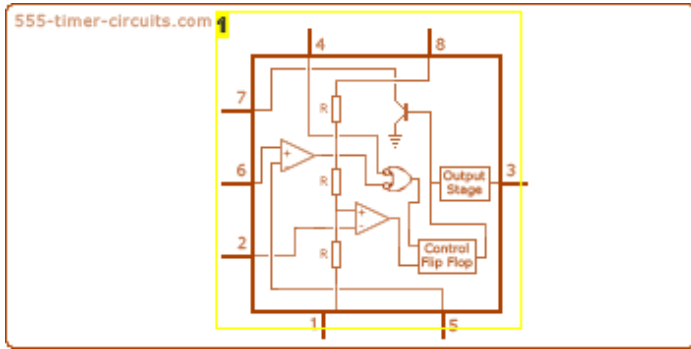


Image Notes

1. This illustration shows the functional block diagram of the 555 timer IC.

Step 4: 555 timer operating modes

The 555 has three main operating modes, Monostable, Astable, and Bistable. Each mode represents a different type of circuit that has a particular output.

Astable mode

An astable circuit has no stable state - hence the name "astable". The output continually switches state between high and low without without any intervention from the user, called a 'square' wave. This type of circuit could be used to give a mechanism intermittent motion by switching a motor on and off at regular intervals. It can also be used to flash lamps and LEDs, and is useful as a 'clock' pulse for other digital ICs and circuits. * refer to pictures one and two

Monostable mode

A monostable circuit produces one pulse of a set length in response to a trigger input such as a push button. The output of the circuit stays in the low state until there is a trigger input, hence the name "monostable" meaning "one stable state". his type of circuit is ideal for use in a "push to operate" system for a model displayed at exhibitions. A visitor can push a button to start a model's mechanism moving, and the mechanism will automatically switch off after a set time. *refer to picture three and four

Bistable Mode (or Schmitt Trigger)

A bistable mode or what is sometimes called a Schmitt Trigger, has two stable states, high and low. Taking the Trigger input low makes the output of the circuit go into the high state. Taking the Reset input low makes the output of the circuit go into the low state. This type of circuit is ideal for use in an automated model railway system where the train is required to run back and forth over the same piece of track. A push button (or reed switch with a magnet on the underside of the train) would be placed at each end of the track so that when one is hit by the train, it will either trigger or reset the bistable. The output of the 555 would control a DPDT relay which would be wired as a reversing switch to reverse the direction of current to the track, thereby reversing the direction of the train. *refer to pictures five and six

<http://www.555-timer-circuits.com/operating-modes.html>

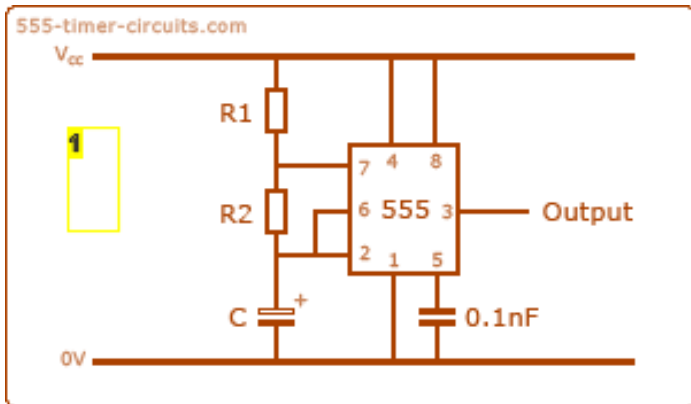


Image Notes

1. picture one-Astable

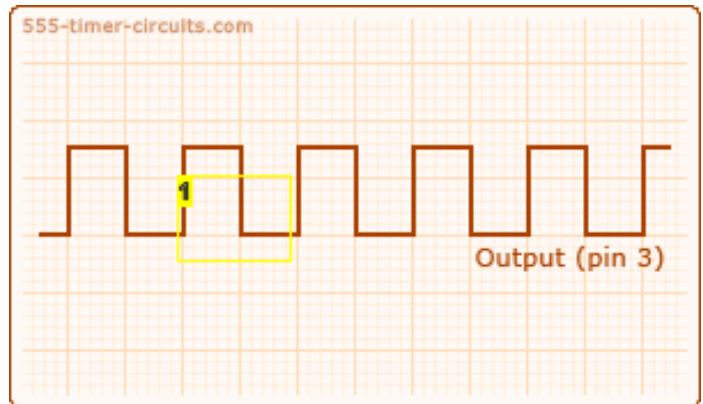


Image Notes

1. picture two-Astable

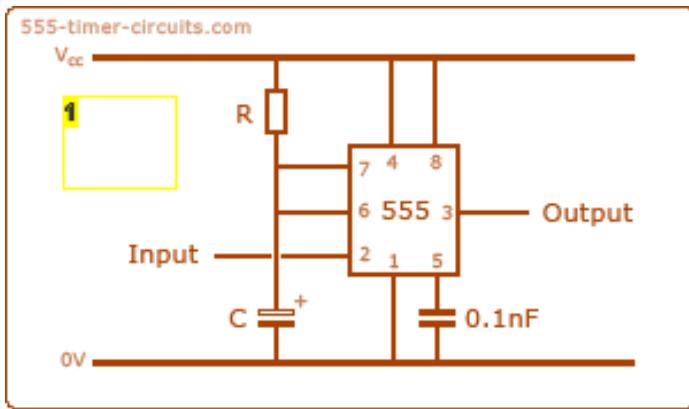


Image Notes

1. picture three-monostable

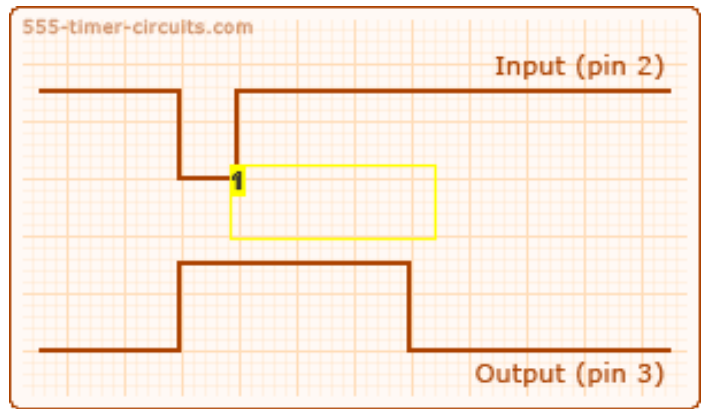


Image Notes

1. picture four-monostable

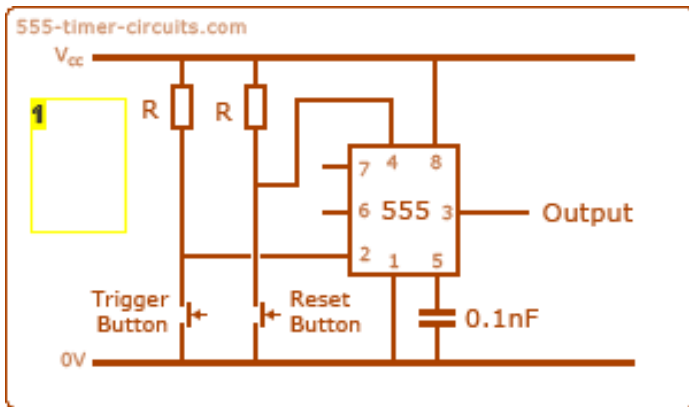


Image Notes

1. picture five-bistable

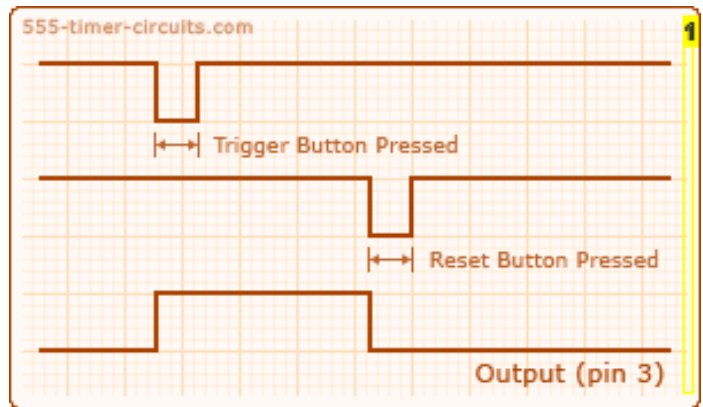


Image Notes

1. picture six-bistable

Step 5: Using the output of a 555 timer

The output (Pin 3) of the 555 can be in one of two states at any time, which means it is a digital output. It can be connected directly to the inputs of other digital ICs, or it can control other devices with the help of a few extra components. The first state is the 'low' state, which is the voltage 0V at the power supply. The second state is the 'high' state, which is the voltage V_{cc} at the power supply.

Sinking and Sourcing

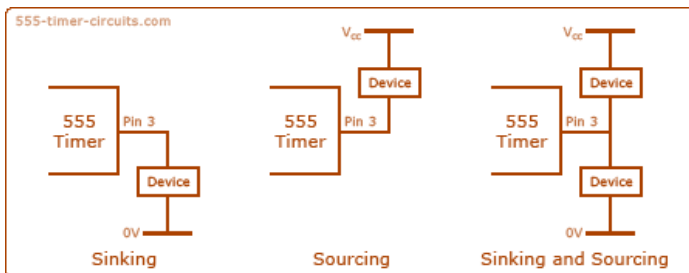
When the Output goes low, current will flow through the device and switch it on. This is called 'sinking' current because the current is sourced from V_s and flows through the device and the 555 to 0V.

When the Output goes high, current will flow through the device and switch it on. This is called 'sourcing' current because the current is sourced from the 555 and flows through the device to 0V.

Sinking and sourcing can also be used together so that two devices can be alternately switched on and off.

The device(s) could be anything that can be switched on and off, such as LEDs, lamps, relays, motors or electromagnets. Unfortunately, these devices have to be connected to the Output in different ways because the Output of the 555 can only source or sink a current of up to 200mA. Make sure your power supply can provide enough current for both the device and the 555, otherwise the timing of the 555 will be affected.

<http://www.555-timer-circuits.com/using-the-output.html>



Step 6: Common mistakes when using a 555 timer

*each numbered step will be in order with the pictures.

1. Pin 7 gets connected to the 0v rail via a transistor inside the chip during part of the operation of the 555. If the pot is turned to very low resistance in the following circuit, a high current will flow through the pot and it will be damaged:

2. The impedance of the 100u electrolytic will allow a very high current to flow and the chip will get very hot. Use 10u maximum when using 8R speaker.

3. The reset pin (pin 4) is internally tied HIGH via approx 100k but it should not be left floating as stray pulses may reset the chip.

4. Do not draw 555 circuits as shown in the following diagram. Keep to a standard layout so the circuit is easy to follow.

5. Here's an example from the web. It takes a lot of time to work out what the circuit is doing:

The aim is to lay-out a circuit so that it shows instantly what is happening. That's why everything must be in recognised locations.

Here is the corrected circuit: From this diagram it is obvious the circuit is an oscillator (and not a one-shot etc).

6. Don't use high value electrolytics and high resistances to produce long delays. The 555 is very unreliable with timing values above 5-10 minutes. The reason is simple. The charging current for the electrolytic is between 1 - 3 microamp in the following diagram (when the electro is beginning to charge) and drops to less than 1 microamp when the electro is nearly charged.

If the leakage of the electro is 1 microamp, it will never fully charge and allow the 555 to "time-out."

7. Do not connect a PNP to the output of a 555 as shown in the following diagram. Pin 3 does not rise high enough to turn off the transistor and the current taken by the circuit will be excessive. Use an NPN driver.

<http://www.555-timer-circuits.com/common-mistakes.html>

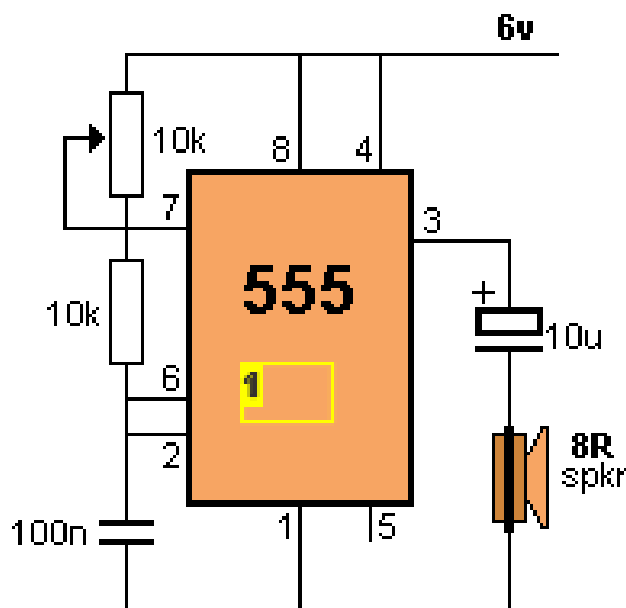


Image Notes

1. step 1

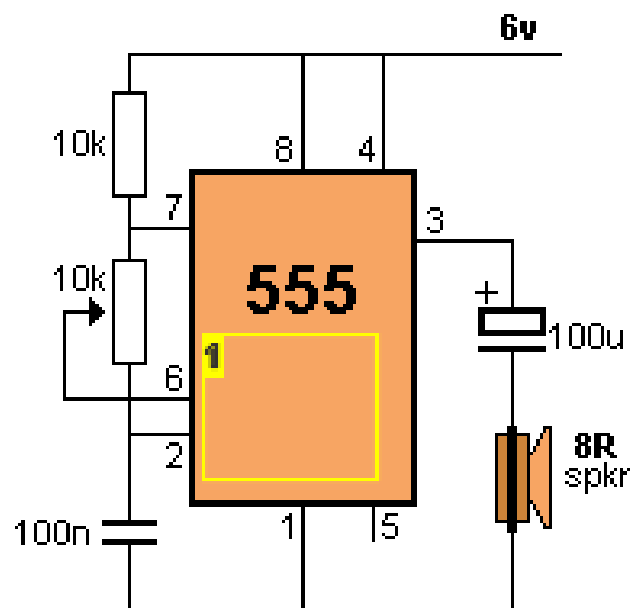
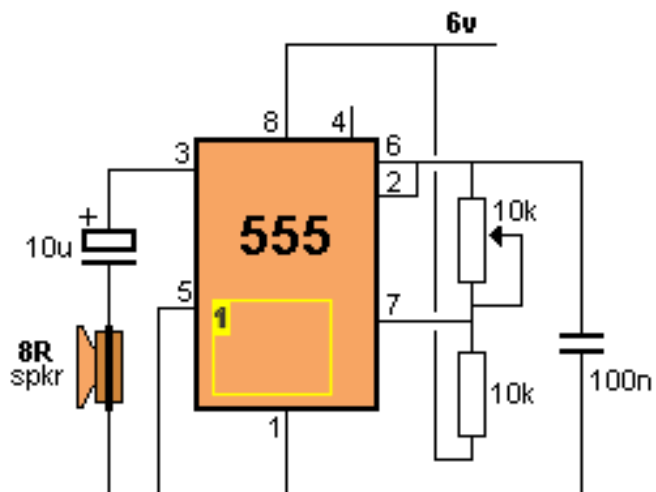


Image Notes

1. step 2



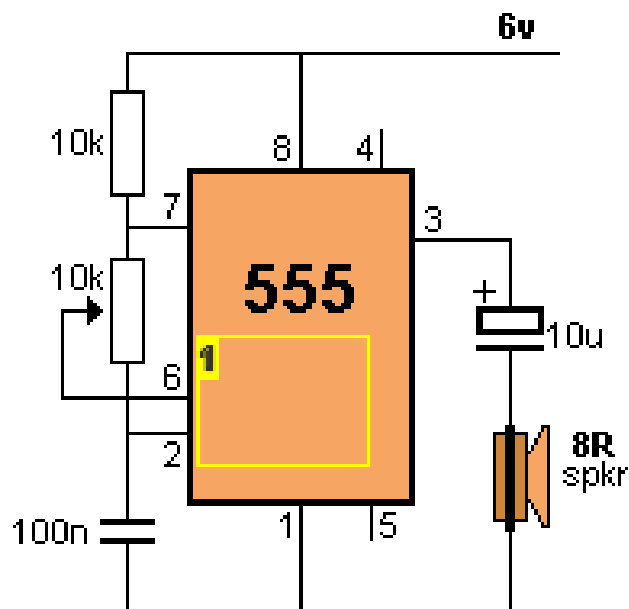


Image Notes
1. step 3

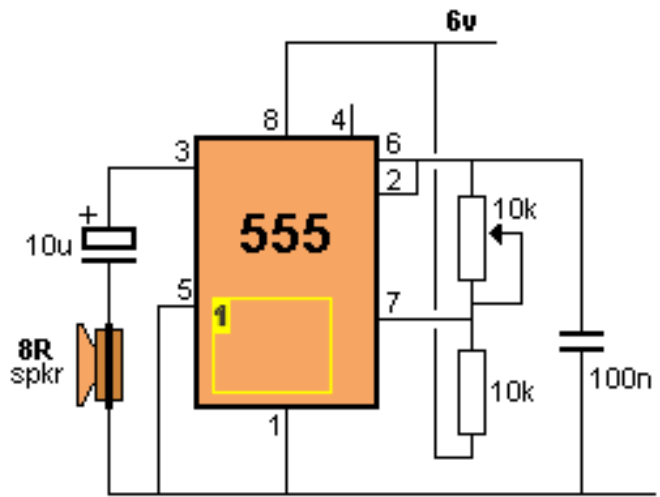


Image Notes
1. step 4

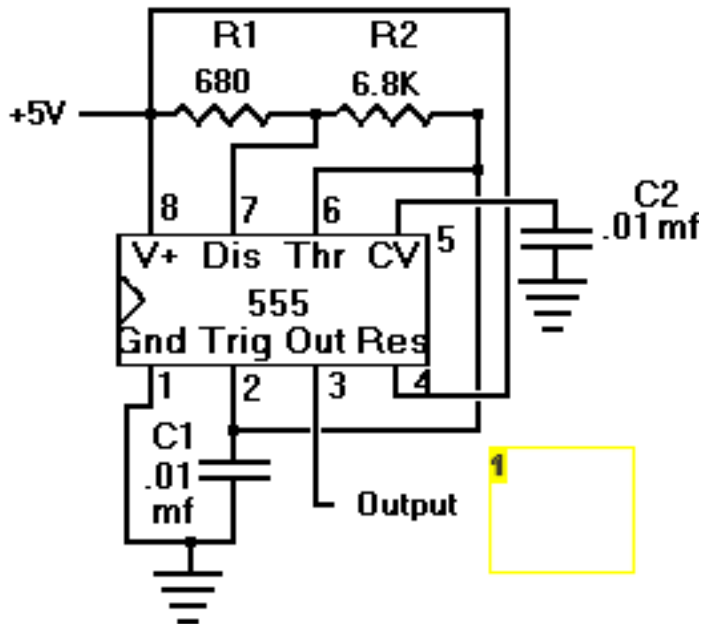


Image Notes
1. step 5

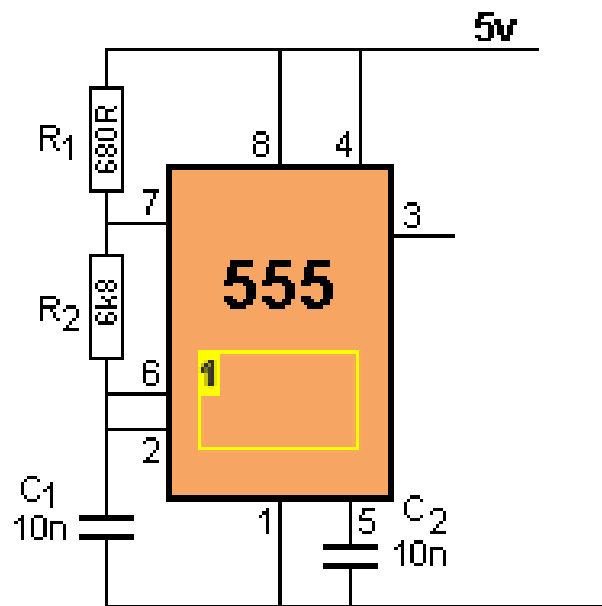


Image Notes
1. step 5

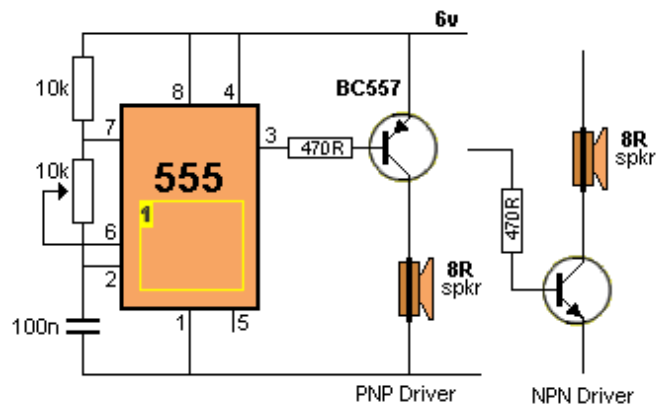
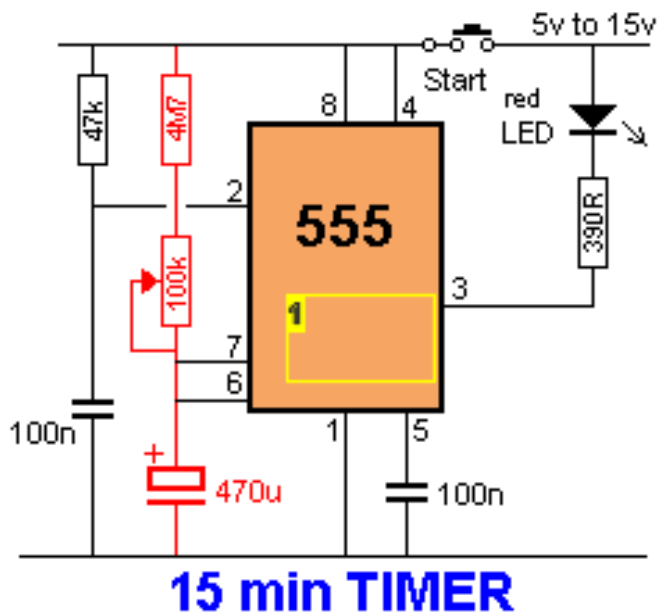


Image Notes

1. step 7

Image Notes

1. step 6



Jcfowler says:

http://www.ehobbycorner.com/pages/tut_timer.html

Thanks man. I found this link to be very helpful as well if you're still blurry on it.

Jan 7, 2011. 2:34 PM [REPLY](#)



nutsandbolts_64 says:

I guess my hand at electronics isn't that good. First off, what is a 0v power supply? Plus, you may want to illustrate how (in step 2, description of pin 7) is, because it my mind it's jumping up and down from several terms in electronics. In short, I get confused at that part, so please do add some form of illustration.

Dec 30, 2010. 3:10 AM [REPLY](#)



Jimmy Proton says:

0v is negative although it should be positive because electrons go from negative to positive (its talking about a DC power supply). and i dont know the answer to that question.

Dec 30, 2010. 1:36 PM [REPLY](#)



nutsandbolts_64 says:

Oh... Once I get what you mean by the aforementioned description of pin 7, I'll *attempt* to make an illustration of how it works.

Dec 30, 2010. 10:06 PM [REPLY](#)



Jimmy Proton says:

I didn't make this stuff up so I dont know what all of it means.

Dec 31, 2010. 12:41 PM [REPLY](#)



nutsandbolts_64 says:

Oh.... OH, right, I forgot, you got the info from another source...

Dec 31, 2010. 11:00 PM [REPLY](#)



Jimmy Proton says:

if you go to the link there's a contact page you ask it on.

Jan 1, 2011. 1:41 PM [REPLY](#)



AndyGadget says:

Jimmy, this is an extremely good guide to the 555 but has been lifted from another website, as you partially acknowledge at the end. This guide also appears in other places on the web, but for it not to be in breach of copyright you need the permission of the original author, either personally or in his general statement of copyright terms. Not doing so is against section 8f of Instructables rules.

Dec 26, 2010. 8:59 AM [REPLY](#)



Jimmy Proton says:

I have permission from the author.

Dec 26, 2010. 11:11 AM [REPLY](#)



xtank5 says:

You may want to detail that in the instructable itself so that it doesn't get pulled when this comment thread is drowned amongst the others that will surely come.

Dec 26, 2010. 11:47 AM [REPLY](#)



AndyGadget says:

Yep, put a credit to the author in the Instructable - Something along the lines of "Material used by permission of xxxxxx with content from XYZ.com."

It's always good to give credit where credit's due, and to tie up any potential ownership disputes before they arise. Keep up the good work.

Dec 26, 2010. 1:39 PM [REPLY](#)